Q1 - 25 July - Shift 1

If the sum and the product of mean and variance of a binomial distribution are 24 and 128 respectively, then the probability of one or two successes is :

(A) $\frac{33}{2^{32}}$ (B) $\frac{33}{2^{29}}$ (C) $\frac{33}{2^{28}}$ (D) $\frac{33}{2^{27}}$

Space for your notes:

Q2 - 25 July - Shift 2

If A and B are two events such that $P(A)=\frac{1}{3}, P(B)=\frac{1}{5}$ and $P(A\cup B)=\frac{1}{2}$, then $P(A|B')+P(B|A')$ is equal to

(A) $\frac{3}{4}$ (B) $\frac{5}{8}$

(C) $\frac{5}{4}$ (D) $\frac{7}{8}$

Space for your notes:

Q3 - 26 July - Shift 1

The mean and variance of a binomial distribution are α and $\frac{\alpha}{3}$ respectively. If $P(X=1)=\frac{4}{243}$, then $P(X = 4 \text{ or } 5)$ is equal to :

(A) $\frac{5}{9}$ (B) $\frac{64}{81}$

(C) $\frac{16}{27}$ (D) $\frac{145}{243}$

Space for your notes:

Q4 - 26 July - Shift 1

Let E_1, E_2, E_3 be three mutually exclusive events such that $P(E_1) = \frac{2+3p}{6}$, $P(E_2) = \frac{2-p}{8}$ and $P(E_3) = \frac{1-p}{2}$. If the maximum and minimum values of p are p_1 and p_2, then $(p_1 + p_2)$ is equal to :

(A) $\frac{2}{3}$ (B) $\frac{5}{3}$

(C) $\frac{5}{4}$ (D) 1

Space for your notes:

Q5 - 26 July - Shift 2

Let X be a binomially distributed random variable with mean 4 and variance $\frac{4}{3}$. Then $54\, P(X \le 2)$ is equal to

(A) $\frac{73}{27}$ (B) $\frac{146}{27}$

(C) $\frac{146}{81}$ (D) $\frac{126}{81}$

Space for your notes:

Q6 - 27 July - Shift 2

Let X have a binomial distribution B(n, p) such that the sum and the product of the mean and variance of X are 24 and 128 respectively. If $P(X > n-3) = \frac{k}{2^n}$, then k is equal to

(A) 528 (B) 529 (C) 629 (D) 630

Space for your notes:

Q7 - 27 July - Shift 2

A six faced die is biased such that $3 \times P(\text{a prime number}) = 6 \times P(\text{a composite number}) = 2 \times P(1)$. Let X be a random variable that counts the number of times one gets a perfect square on some throws of this die. If the die is thrown twice, then the mean of X is :

(A) $\frac{3}{11}$ (B) $\frac{5}{11}$

(C) $\frac{7}{11}$ (D) $\frac{8}{11}$

Space for your notes:

Q8 - 28 July - Shift 1

Out of 60% female and 40% male candidates appearing in an exam, 60% candidates qualify it. The number of females qualifying the exam is twice the number of males qualifying it. A candidate is randomly chosen from the qualified candidates. The probability, that the chosen candidate is a female, is :

(A) $\frac{2}{3}$ (B) $\frac{11}{16}$

(C) $\frac{23}{32}$ (D) $\frac{13}{16}$

Space for your notes:

Q9 - 28 July - Shift 2

Let A and B be two events such that $P(B|A)=\frac{2}{5}$, $P(A|B)=\frac{1}{7}$ and $P(A\cap B)=\frac{1}{9}$. Consider

$(S1)\,P(A'\cup B)=\frac{5}{6}$,

$(S2)\,P(A'\cap B')=\frac{1}{18}$. Then

(A) Both (S1) and (S2) are true

(B) Both (S1) and (S2) are false

(C) Only (S1) is true

(D) Only (S2) is true

Space for your notes:

Q10 - 28 July - Shift 2

A bag contains 4 white and 6 black balls. Three balls are drawn at random from the bag. Let X be the number of white balls, among the drawn balls. If σ^2 is the variance of X, then $100\,\sigma^2$ is equal to

Space for your notes:

Q11 - 29 July - Shift 1

Let S = {1, 2, 3, …, 2022}. Then the probability, that a randomly chosen number n from the set S such that HCF (n, 2022) = 1, is :

(A) $\frac{128}{1011}$ (B) $\frac{166}{1011}$

(C) $\frac{127}{337}$ (D) $\frac{112}{337}$

Space for your notes:

Q12 - 29 July - Shift 2

Bag I contains 3 red, 4 black and 3 white balls and Bag II contains 2 red, 5 black and 2 white balls. One ball is transferred from Bag I to Bag II and then a ball is draw from Bag II. The ball so drawn is found to be black in colour. Then the probability, that the transferred ball is red, is:

(A) $\frac{4}{9}$ (B) $\frac{5}{18}$ (C) $\frac{1}{6}$ (D) $\frac{3}{10}$

Space for your notes:

Q13 - 29 July - Shift 2

The sum and product of the mean and variance of a binomial distribution are 82.5 and 1350 respectively. They the number of trials in the binomial distribution is:

Space for your notes:

Answer Key

Q1 (C)	**Q2** (B)	**Q3** (C)	**Q4** (D)
Q5 (B)	**Q6** (B)	**Q7** (D)	**Q8** (A)
Q9 (A)	**Q10** (56)	**Q11** (D)	**Q12** (B)
Q13 (96)			

Q1 (C)

$np + npq = 24 \quad \ldots(1)$

$np \cdot npq = 128 \quad \ldots(2)$

Solving (1) and (2) :

We get $p = \frac{1}{2}, q = \frac{1}{2}, n = 32.$

Now,

$P(X = 1) + P(X = 2)$

$= {}^{32}C_1\, pq^{31} + {}^{32}C_2\, p^2 q^{30}$

$= \frac{33}{2^{28}}$

Q2 (B)

(B)

Q3 (C)

$np = \alpha$ .....(1)

$npq = \alpha /3$ .....(2)

From (1) & (2)

$q = 1/3$ & $p = 2/3$

$$^{n}C_{1}\, q^{n-1} p^{1} = \frac{4}{243}$$

$$\frac{n}{3^{n}} = \frac{2}{243}$$

$n = 6$

$$P\,(4 \text{ or } 5) = {}^{6}C_{4}\left(\frac{2}{3}\right)^{4}\left(\frac{1}{3}\right)^{2} + {}^{6}C_{5}\left(\frac{2}{3}\right)^{5}\cdot\left(\frac{1}{3}\right)^{0}$$

$$= \frac{16}{27}$$

Q4 (D)

$0 \le P(E_i) \le 1$ for $i = 1, 2, 3$

$\Rightarrow -2/3 \le p \le 1$

E_1 & E_2 & E_3 are mutually exclusive

$P(E_1) + P(E_2) + P(E_3) \le 1$

$\Rightarrow 2/3 \le p \le 1$

$p_1 = 1,\ p_2 = 2/3$

$p_1 + p_2 = 5/3$

Q5 (B)

$np = 4$

$npq = 4/3$

$n = 6,\ p = 2/3,\ q = 1/3$

$54(P(X = 2) + P(X = 1) + P(X = 0))$

$$54\left({}^6C_2\left(\frac{2}{3}\right)^2\left(\frac{1}{3}\right)^4 + {}^6C_1\left(\frac{2}{3}\right)^1\left(\frac{1}{3}\right)^5 + {}^6C_0\left(\frac{2}{3}\right)^0\left(\frac{1}{3}\right)^6\right)$$

$$= \frac{146}{27}$$

Q6 (B)

Let α = Mean & β = Variance $(\alpha > \beta)$

So, $\alpha + \beta = 24$, $\alpha\beta = 128$

$\Rightarrow \alpha = 16$ & $\beta = 8$

$\Rightarrow np = 16$ $\quad npq = 8 \Rightarrow q = \frac{1}{2}$

$$\therefore\ p = \frac{1}{2},\ n = 32$$

$$p(x > n-3) = \frac{1}{2^n}\left({}^nC_{n-2} + {}^nC_{n-1} + {}^nC_n\right)$$

$$\therefore k = {}^{32}C_{30} + {}^{32}C_{31} + {}^{32}C_{32} = \frac{32 \times 31}{2} + 32 + 1$$

$$= 496 + 33 = 529$$

Q7 (D)

Let $\frac{P(\text{a prime number})}{2} = \frac{P(\text{a composite})}{1} = \frac{P(1)}{3} = k$

So, $P(\text{a prime number}) = 2k$,

$P(\text{a composite number}) = k$,

& $P(1) = 3k$

& $3 \times 2k + 2 \times k + 3k = 1$

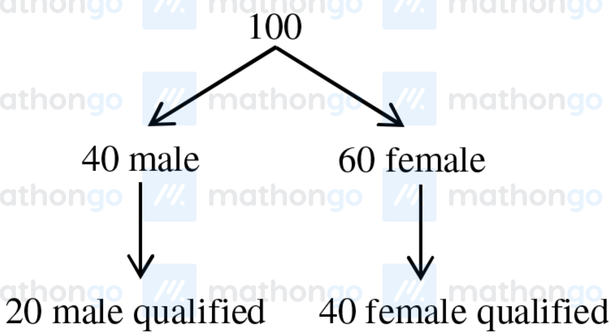
$\Rightarrow k = \frac{1}{11}$

$P(\text{success}) = P(1 \text{ or } 4) = 3k + k = \frac{4}{11}$

Number of trials, $n = 2$

$\therefore$ mean $= np = 2 \times \frac{4}{11} = \frac{8}{11}$

Q8 (A)



Probability that chosen candidate is female $= \frac{40}{60} = \frac{2}{3}$

Q9 (A)

$$P(A|B)=\frac{1}{7}\Rightarrow\frac{P(A\cap B)}{P(B)}=\frac{1}{7}$$

$$\Rightarrow P(B)=\frac{7}{9}$$

$$P(B|A)=\frac{2}{5}\Rightarrow\frac{P(A\cap B)}{P(A)}=\frac{2}{5}$$

$$\Rightarrow P(A)=\frac{5}{18}$$

Now, $P(A'\cup B)=1-P(A\cup B)+P(B)$

$$=1-P(A)+P(A\cap B)\ =\frac{5}{6}$$

$$P(A'\cap B')=1-P(A\cup B)$$

$$=1-P(A)-P(B)+P(A\cap B)=\frac{1}{18}$$

$\Rightarrow$ Both (S1) and (S2) are true.

Q10 (56)

X	0	1	2	3
$P(X)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{3}{10}$	$\frac{1}{30}$

$$\sigma^2=\sum X^2P(X)-\left(\sum XP(X)\right)^2=\frac{56}{100}$$

$$100\ \sigma^2=56$$

Q11 (D)

Total number of elements = 2022

$2022 = 2 \times 3 \times 337$

HCF (n, 2022) = 1

is feasible when the value of 'n' and 2022 has no common factor.

A = Number which are divisible by 2 from $\{1,2,3.....2022\}$

$n(A) = 1011$

B = Number which are divisible by 3 by 3 from $\{1,2,3......2022\}$

$n(B) = 674$

$A \cap B =$ Number which are divisible by 6 from $\{1,2,3........2022\}$

6,12,18........., 2022

$$\boxed{337 = n(A \cap B)}$$

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$= 1011 + 674 - 337$

$= 1348$

C= Number which divisible by 337 from $\{1,........1022\}$

C= {337,674,1011,1348,1685,20222}

Already counted in Set $(A \cup B)$

Already counted in Set $(A \cup B)$

Already counted in Set $(A \cup B)$

Total elements which are divisible by 2 or 3 or 337

$= 1348 + 2 = 1350$

Favourable cases = Element which are neither divisible by 2, 3 or 337

$= 2022 - 1350$

$= 672$

Required probability = $\frac{\quad}{2022} = \frac{\quad}{337}$

Q12 (B)

I	II
3R	2R
4B	5B
3W	2W

A : Drown ball from boy II is black

B : Red ball transferred

$$P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)}$$

$$= \frac{\frac{3}{9} \times \frac{5}{10}}{\frac{3}{9} \times \frac{5}{10} + \frac{4}{9} \times \frac{6}{10} + \frac{3}{9} \times \frac{5}{10}}$$

$$= \frac{15}{15 + 24 + 15} = \frac{15}{54} = \frac{5}{18}$$

Q13 (96)

Let, mean $= m = np$

& variance $= v = npq$, $p + q = 1$

Sum $= m + v = \dfrac{165}{2}$

Product $= mv = 1350$

On solving,

$m = np = 60$ & $v = npq = \dfrac{45}{2}$ $\therefore$ $q = \dfrac{3}{8}$ $\therefore$ $P = \dfrac{5}{8}$

Hence $n = 96$